BU-RSI

ROTOR-STATOR INTERACTION PROGRAM

USER DOCUMENTATION

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Compilation of the fortran codes requires access to compiled libraries: nasalib and linpacklib which are included with the distribution of the code. They may need to be recompiled. There is a Make file in each directory.

FREQUENCY Check Module (in FINDFREQS directory)

Executable:

comp2

Input File

areo95const.input — Input file in Namelist format (described below) , you can link this to the file in the EXAMPLE directory

Output file

fort.102 — 3 column ascii (circumferential mode number, radial mode number, eigen frequency for annulus)

Makefile

MakeComp2 (which runs Makecomp2.pgi)

Run

Type comp2 (note at the end you may get an IEEE warning – ignore)

freqs2.m **matlab code – follow up to comp2**

may need to be manually corrected to get good distribution of frequencies

Input :

M_ann, Rtip need to be set in the freqs2.m file, fort.102 which is generated by comp2 must be in the directory

Output files:

Fs.sing — singular frequencies for acoustic propagation in annulus

Fs.dat — non-singular frequencies for acoustic propagation in annulus

User will want to edit the fs.dat file to select a suitable number of frequencies to use in the calculation.

RSI (In CODE directory)

Executable:

rsi95

Input Files:

aero95const.input	Input file in Namelist format
aero95mat.input	Input file in Namelist format
turb95mat.input	Input file in Namelist format
fs.dat	single column ascii (can be generated using CompXmn2.f95 and freqs2.m)

Output File of interest:

fort.33	Exhaust duct spectrum

Namelist Input for Input File "aero95const.input":

NV = *Number of vanes in Stator*

NB = *Number of blades in Rotor*

NRdat = *Number of spanwise strips at which input is specified*

Number of spanwise positions on which input will be specified, including endwalls.

NRcalc = *Number of spanwise strips at which vane lift response is to be calculated*

Spanwise positions are equally-spaced, and are independent of the NRdat positions. Values that are input on NRdat spanwise positions will be interpolated onto NRcalc spanwise positions. *SHOULD BE SET BY RADIAL TURBULENT LENGTH SCALE -- strips should be roughly equivalent to value of radial length scale*

NC = *Number of chordwise points at which vane lift response is to be calculated*

S. Grace indicated that the value of NC should always be >= 80, based on the value of k_z. If k_z <= 60, then NC ~= 80, and if k_z > 60, then NC > 80.

NF = *The number of frequencies that will be computed from the list in fs.dat*

Ng = *Number of terms for k3 discretization. K3 currently varies from 0 to 24 in 24/Ng increments. This can be changed by modifying dk3 in the rsi.f95 code*

T = *Averaged temperature inside the duct (in Rankine)*

Static temperature, spanwise mass-averaged at Vane LE. T is used only to compute dB_ref

Currrently – not utilized in the code. Speed of Sound set to 1115 ft/s.

rho = *Averaged density inside the duct (lbm/ft**3)*

Spanwise mass-averaged. Used only for calculation of dB_ref.

M_ann = *Duct bulk Mach number (assumed to be plug flow)*

Mach number based on axial velocity component downstream of Vane TE.

iSTDYL = *Switch to turn on the Incidence Angle correction in the calculation of Mach number normal to the Vane LE (not currently used)*

This parameter appears in rsi_aero.f95. If iSTDYL = 1, then the Incidence Angle correction is applied in the calculation of Mach number normal to the Vane LE. If iSTDYL = 0, then the correction is not applied, i.e., it is assumed that the flow angle is aligned with the stagger angle of the vane.

[S. Grace has not explored the use of this parameter, and has always set iSTDYL = 0.]

tipc_to_tipr = *Ratio of Vane tip chord to duct radius*

Vane tip chord, not axial chord.

Duct radius specified at Vane LE.

hub_to_tip = *Duct hub-to-tip radius ratio*

Computed from values at Vane LE:

$$hub_to_tip = \frac{r(1)}{r(NRdat)}$$

sweepdeg = *Vane sweep angle (degrees)*

Sweep angle at Vane LE. Sweep angle measured from radial, with aft sweep > 0.

RDUCT = Duct radius at stator leading edge (inches) – used for nondimensionalization when needed

BW = *Spectral bandwidth (Hz)*

Used to compute bandwidth correction, BWc, for use in Power calculations, along with dB_ref.

$$BWc = 10 \cdot \log_{10}(BW)$$

[S. Grace indicates that this input parameter was used for purposes of matching test data bandwidth. If a calculation is based on prediction, rather than matching with test data, set BW = 0.]

Conv_tol_xmn = *Root finding tolerance limit*

Identified as convergence criterion for calculation of the nondimensional root of the Eigenvalue equation.

[Example input file uses ec = 0.0001. S. Grace has always used this value.]

Namelist Input for Input File "aero95mat.input":

radius = *Radii at which data are specified (non-dimensionalized by tip radius)*

Array (1:NRdat)

Computed from values at Vane LE:

$$radius_i = \frac{r_i}{r_{NRdat}}$$

chord = *Vane chord at each radius (non-dimensionalized by tip chord)*

Array (1:NRdat)

Vane tip chord, not axial chord.

Computed from values for Vane:

$$chord_i = \frac{Chord_i}{Chord_{NRdat}}$$

stagger = *Vane stagger angle at each radius (degrees)*

Array (1:NRdat)

Angle measured from axial direction. Stagger angle > 0, to be consistent with RPM > 0.

Computed from values for Vane:

$$stagger(i) = -alpha(i) \cdot \left(\frac{180.}{\pi} \right)$$

Mx = *Mean absolute axial Mach number at each radius (at Vane leading edge)*

Array (1:NRdat)

Mach number based on axial component of velocity.

Mt = *Mean absolute tangential Mach number at each radius (at Vane leading edge)*

Array (1:NRdat)

Mach number based on tangential component of velocity.

Namelist Input for Input File "turb95mat.input":

The following turbulence quantities may be available from a CFD analysis of the Fan stage, or from hot wire or laser velocimetry measurements conducted during a test. For those cases where no CFD or measured data is available (e.g., for a new Fan design), it is recommended to use the turbulence information from the CFD analysis of a similar Fan stage configuration, in conjunction with the assumptions described below.

CURRENTLY – the code uses only an average passage value of turbulence intensity. So maxurms and urmsgausswidth are set to zero and backurms is taken as the average passage value of the turbulence intensity. (The other two parameters are artifacts left from when a Guassian fit was made to the TKE data and then the turbulence intensity was calculated from TKE.)

backurms = *Average passage value of RMS turbulence intensity at each radius (non-dimensionalized by mean absolute velocity at that radius)*

Array (1:NRdat)

Computed from CFD solution or experimental data at/upstream of Vane LE:

maxurms = *not used, set to zero*

urmsguasswidth = *not used, set to zero*

Lxdat = *Longitudinal integral length scale (non-dimensionalized by duct tip radius)*

Array (1:NRdat)

Longitudinal integral length scale is defined along the Vane chord, not in the axial direction.

Computed from CFD solution or experimental data at/upstream of Vane LE:

Ltdat = *Upwash integral length scale (non-dimensionalized by duct tip radius)*

Array (1:NRdat)

Upwash integral length scale is defined normal to the Vane chord, not in the tangential direction.

Computed from CFD solution or experimental data at/upstream of Vane LE:

For CFD solution, assume isotropic turbulence, with $L_t = L_x/2$

Lrdat = *Radial integral length scale (non-dimensionalized by duct tip radius)*

Array (1:NRdat)

Radial integral length scale is defined in radial direction.

Computed from CFD solution or experimental data at/upstream of Vane LE:

For CFD solution, assume isotropic turbulence, with $L_r = L_x/2$

Output File of interest "fort.33": example format

Frequency (Hz)	Inlet Power (dB)	Exhaust Power (dB)
2320.45000000000	95.8128864000951	102.416565511673

While Inlet power is output from the code, this part of the calculation has never been validated. Only exhaust values have been shown to be correct.

Note: In rsi.f95, the Power Density calculation is:

$$sPWR = 10 \cdot \log_{10}\left(sPWR\right) + dB_ref + BWc$$

where dB_ref takes care of making things dimensional and BWc is the bandwidth correction based on experimental data collection method. I usually don't enter BWc in the input. I adjust afterwards.